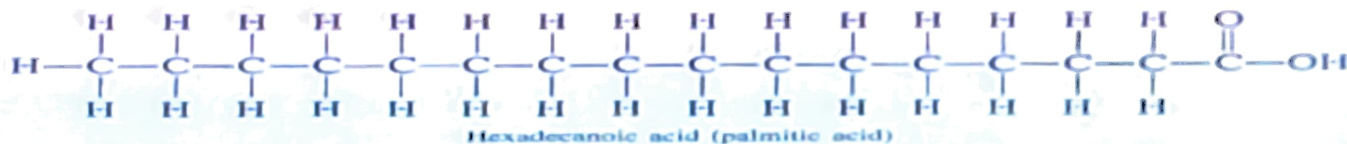
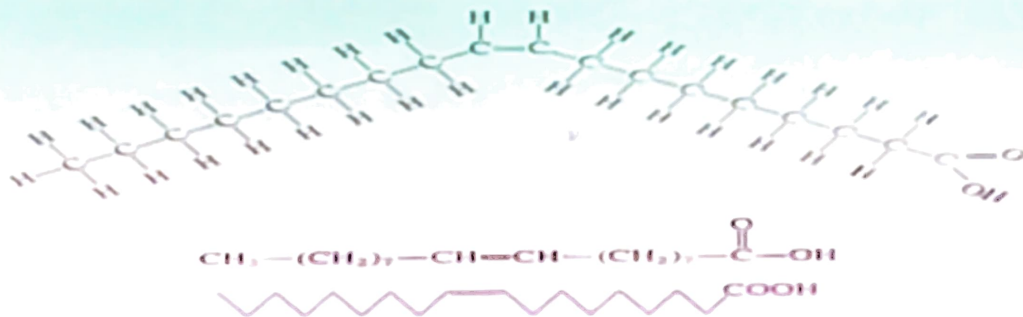


Fatty acids

➤ **Saturated fatty acid**: Fatty acid chains that contain only carbon-carbon single bonds are referred to as saturated (Palmitic acid)

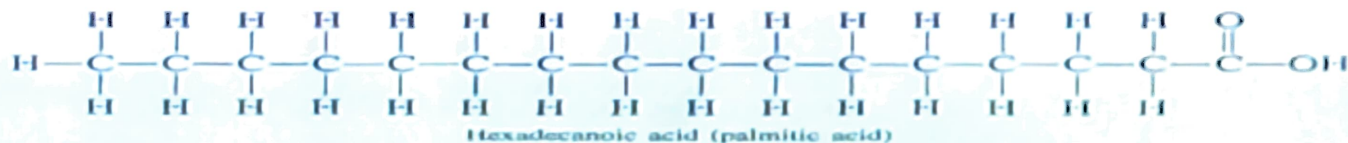


➤ **Unsaturated fatty acid**: Those molecules that contain one or more double bonds are said to be unsaturated. There are mono- and polyunsaturated fatty acids (Oleic acid)

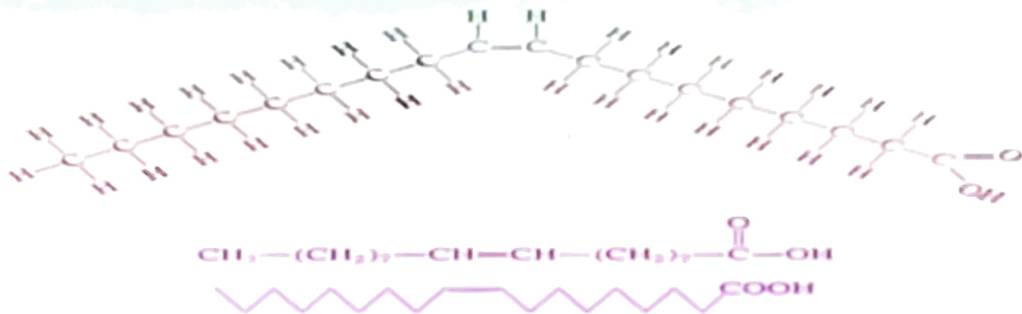


Fatty acids

➤ **Saturated fatty acid**: Fatty acid chains that contain only carbon-carbon single bonds are referred to as **saturated** (Palmitic acid)



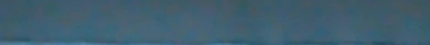
➤ **Unsaturated fatty acid**: Those molecules that contain one or more double bonds are said to be unsaturated. There are mono- and polyunsaturated fatty acids (Oleic acid)



Fatty acids

- **Fatty acids** are naturally occurring carboxylic acids with an unbranched carbon chain and an even number of carbon atoms.
- The pathway by which fatty acids are biosynthesized they almost always contain an even number of carbon atoms.
- Long-chain fatty acids (12 to 26 carbon atoms) are found in meats and fish
- Medium-chain fatty acids (6 to 10 carbon atoms) and short-chain fatty acids (fewer than 6 carbon atoms) occur primarily in dairy products.

There are saturated and unsaturated Fatty acids.

Number of carbons	Common name	Systematic name	Structure	point °C
Saturated				
12	lauric acid	dodecanoic acid		44
14	myristic acid	tetradecanoic acid		58
16	palmitic acid	hexadecanoic acid		63
18	stearic acid	octadecanoic acid		69
20	arachidic acid	eicosanoic acid		77
Unsaturated				
16	palmitoleic acid	(9Z)-hexadecenoic acid		0
18	oleic acid	(9Z)-octadecenoic acid		13
18	linoleic acid	(9Z,12Z)-octadecadienoic acid		-5
18	linolenic acid	(9Z,12Z,15Z)-octadecatrienoic acid		-11
20	arachidonic acid	(5Z,8Z,11Z,14Z)-eicosatetraenoic acid		-50
20	EPA	(5Z,8Z,11Z,14Z,17Z)-eicosapentaenoic acid		-50

Number of carbons	Common name	Systematic name	Structure	melting point °C
Saturated				
12	lauric acid	dodecanoic acid		-44
14	myristic acid	tetradecanoic acid		58
16	palmitic acid	hexadecanoic acid		63
18	stearic acid	octadecanoic acid		69
20	arachidic acid	eicosanoic acid		77
Unsaturated				
16	palmitoleic acid	(9Z)-hexadecenoic acid		0
18	oleic acid	(9Z)-octadecenoic acid		13
18	linoleic acid	(9Z,12Z)-octadecadienoic acid		-5
18	linolenic acid	(9Z,12Z,15Z)-octadecatrienoic acid		-11
20	arachidonic acid	(5Z,8Z,11Z,14Z)-eicosatetraenoic acid		-50
20	EPA	(5Z,8Z,11Z,14Z,17Z)-eicosapentaenoic acid		-50

What are Essential Fatty Acids (EPA)?

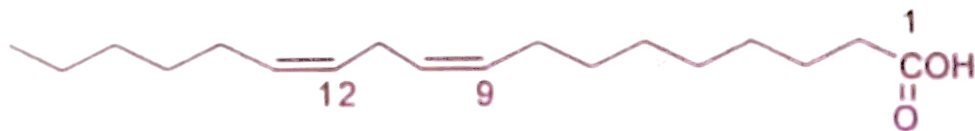
There are TWO essential fatty acids.

Essential means you NEED to get them from the diet because the body cannot manufacture them.

- α -linolenic acid (LNA or ALA): an omega-3 fatty acid

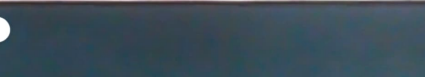


- Linoleic acid (LA): an omega-6 fatty acid found abundantly in soy oil, sunflower seeds, pumpkin seeds, sesame seeds, corn oil, and in most nuts.



Linoleic acid

cis-9,12-Octadecadienoic acid, 18:2($\Delta^{9,12}$)

Number of carbons	Common name	Systematic name	Structure	point °C
Saturated				
12	lauric acid	dodecanoic acid		44
14	myristic acid	tetradecanoic acid		58
16	palmitic acid	hexadecanoic acid		63
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Functions of EFA

- Essential fatty acids are required for the membrane structure and function.
- Transport of cholesterol.
- Formation of lipoproteins.
- Prevention of fatty liver etc.
- They are also needed for the synthesis of another important group of compounds, namely eicosanoids.

Rancidity

Rancidity: Rancidity is the term used to represent the deterioration of fats and oils resulting in an unpleasant taste. Fats containing unsaturated fatty acids are more susceptible to rancidity.

- Rancidity occurs when fats and oils are exposed to air, moisture, light, bacteria etc.
- Hydrolytic rancidity occurs due to partial hydrolysis of triacylglycerols by bacterial enzymes.
- Oxidative rancidity is due to oxidation of unsaturated fatty acids.

This results in the formation of unpleasant products such as dicarboxylic acids, aldehydes, ketones etc.

Rancid fats and oils are unsuitable for human consumption.

Test to Check Purity of Fats and Oils

- **Acid number:-** It is the number of milligrams of potassium hydroxide (KOH) required to completely neutralize the free fatty acids present in 1 g of the oil or fat.
- In normal circumstances refined oils should be free from any free fatty acids.
- Oils, on decomposition due to chemical or bacterial contamination-yield free fatty acids.

Therefore, oils with increased acid number are unsafe for human consumption

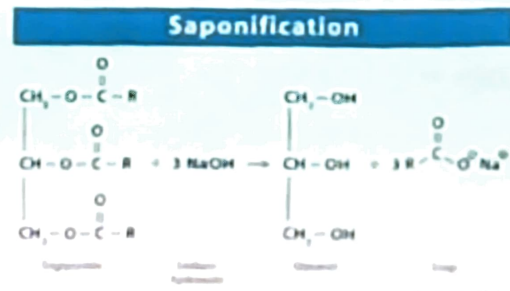
- **Saponification number:-** It is defined as the mg (number) of KOH required to hydrolyze (saponify) one gram of fat or oil.
- Saponification number is a measure of the average molecular size of the fatty acids present
- The value is higher for fats containing short chain fatty acids.

Human fat : 195-200

Butter : 230-240

Coconut oil : 250-260

Saponification is the chemical reaction, specifically alkaline hydrolysis, that converts fats and oils (triglycerides) into glycerol and soap (fatty acid salts)



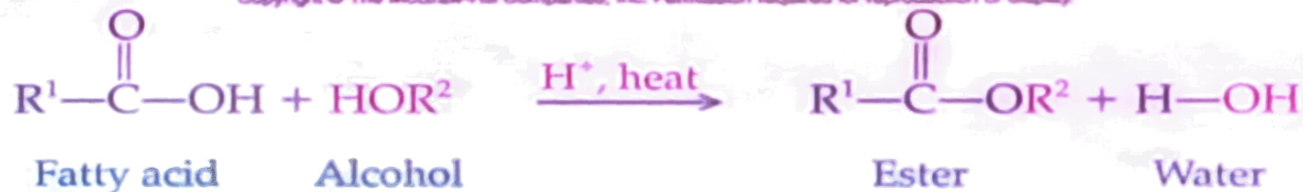
- **Iodine number:-** It is defined as the grams (number) of iodine absorbed by 100 g of fat or oil.
- Iodine number is useful to know the relative unsaturation of fats, and is directly proportional to the content of unsaturated fatty acids.
- Thus lower is the iodine number, less is the degree of unsaturation.

Determination of iodine number will help to know the degree of adulteration of a given oil.

Chemical Reactions of Fatty Acids

Esterification : fatty acids reacts with alcohols to form esters and water

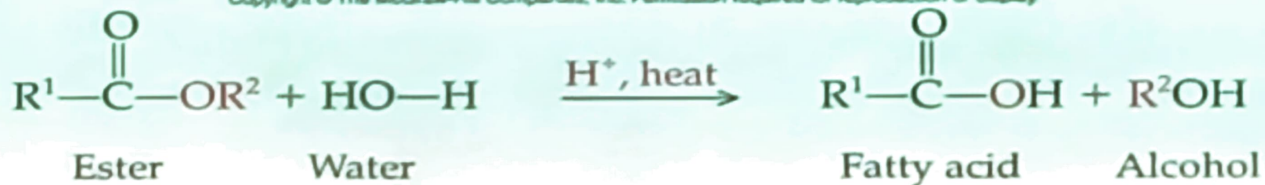
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Fatty Acid Hydrolysis

- Acid Hydrolysis reverses esterification. Fatty acids are produced from esters

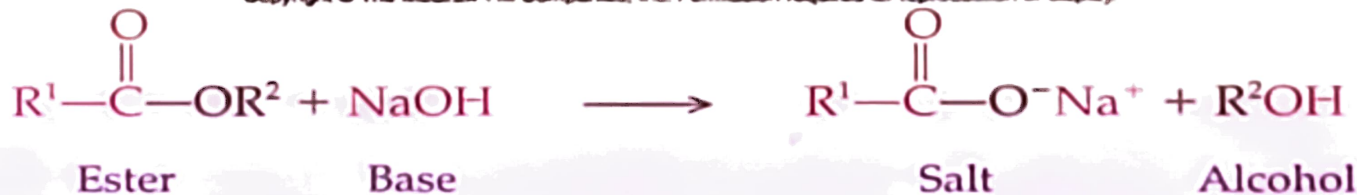
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Saponification

- Saponification is the base-catalyzed hydrolysis of an ester
- Products of the reaction are
 - An alcohol
 - An ionized salt which is a soap
 - Soaps have a long uncharged hydrocarbon tail
 - Also have a negatively charged carboxylate group at end
 - Form micelles that dissolve oil and dirt particles

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